CS 2490 – Homework 6

Turn in by 2026/10/17 at 11:59pm

You **must** turn in your assignment as a single zip file entitled: *lastname_firstname_hw7.zip*. It will contain a single PDF file named *lastname_firstname_hw7.pdf* with your solutions to the exercises, and JSON files for each relevant exercise named *lastname_firstname_hw7_exX.json* where X is the exercise number.

Exercise 1. Consider the grammar $G = (\{a, b\}, \{S\}, S, P)$ where P consists of the following production rules:

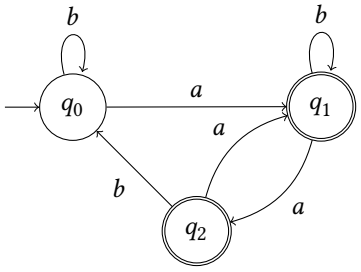
$$S \to aaS \mid a \mid bb$$

(i) Is G a regular grammar? Justify your answer.

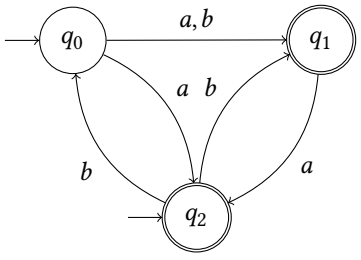
(ii) Construct a NFA that accepts L(G).

Include the JSON file for your NFA in your submission.

Exercise 2. Construct a regular grammar that generates the language accepted by the following DFA:



Exercise 3. Construct a regular grammar that generates the language accepted by the following NFA:



Exercise 4. Construct a regular grammar for the language L over the alphabet $\{0, 1\}$ consisting of all binary representations of multiples of 3.

Hint: don't try to guess the grammar! Use the methods seen in class.

Exercise 5. In class, we have seen right-linear grammars. *Left-linear grammars* are defined similarly: a grammar $G = (\Sigma, V, S, P)$ is left-linear if all its production rules are of the form $A \to Bw$ or $A \to w$ where $A, B \in V$ and $w \in \Sigma^*$.

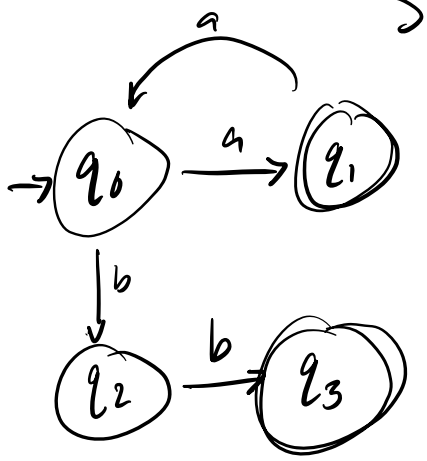
Show that a language is generated by a left-linear grammar if and only if it is generated by a right-linear grammar.

Ex1 1 is G regular

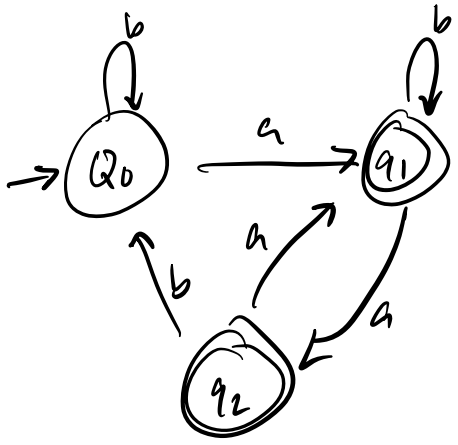
No, because the production rules have more than one non-terminal symbol

$S \to aaS \mid a \mid bb$

11



Ex2

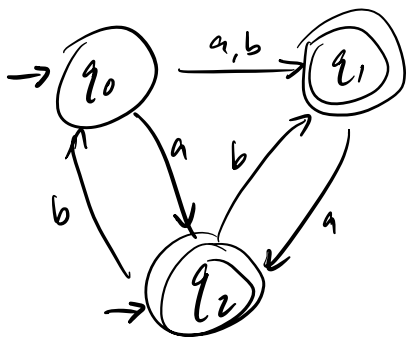


$G = (\{a, b\}, \{S\}, S, P)$

valid
- a
- ba
- baba
- aaba
- aabab

P →
- a
- aS
- bS
- aabaS
- aababS

Ex 3



$G_1 = (\{a, b\}, \{S\}, S, P)$

$P \rightarrow a$
b
aa
baS
bS
abS

Ex 4 $G_1 = (\{0, 1\}, \{S\}, S, P)$

all binary representations of multiples of 3

0011
0110
1001

Production Rules

$A \rightarrow 0A \mid 1B \mid \varepsilon$

$B \rightarrow 0C \mid 1A$

$C \rightarrow 0B \mid 1C$

Ex5

Both left and right-linear languages define what it means to be a regular language. If you can define a right-linear grammar with production rules so that it can be constructed backwards, therefore you could construct a right linear language left-wise. and vice versa.

I don't have a mastery of the vocab... but if it bleeds, we can kill it.